

Jaisidh Singh

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Education

- 2024–present **MSc in Machine Learning**, *Eberhard-Karls Universität Tübingen*, Tübingen.
(expected 2026) GPA: 1.52/5.0 (German scale, 1.0 to 5.0 from best to worst)
Thesis (with OpenEuroLLM: Aaron Klein & Antonio Orvieto): scaling laws for hybrid LLMs
- 2020–2024 **B.Tech in AI & Data Science**, *Indian Institute of Technology*, Jodhpur.
Thesis: negation-aware compositional reasoning in vision-language models.

Work/Research Experience

- Apr 2026 – present **Thesis Researcher at OpenEuroLLM**, *With Dr. Aaron Klein & Dr. Antonio Orvieto*.
 - Improved data efficiency by $\approx 30\%$ in LLM pretraining with hybrid-attention architectures.
 - Developed an efficient & comprehensive backend for LLM scaling laws, including automatic throughput profiling, cost estimation, management & analysis of experiments to **cut total GPU-hours by up to 5×**.
- Oct 2025 – May 2026 **SWE Graduate Assistant at Tübingen AI Center**, *With Prof. Peter Gehler*.
 - Interactive Art Exhibit for Stadtmuseum Tübingen**: Designed & deployed end-to-end real-time interactive computer vision systems for artistic stylization of user images using semantic segmentation and pose estimation based warping.
 - TuebingenAICenter/agent-tutorial**: Designed & delivered a hands-on workshop on LLM-agent orchestration with LangGraph at GCPR 2025, covering tool-use, memory, & MCP towards a scholarly deep-research multi-agent system.
- Jan 2025 – present **ML Research Assistant at ELLIS Institute Tübingen**, *With Dr. Antonio Orvieto*.
 - Reduced VLM training cost by 90%** by using a hypernetwork-based framework for modality stitching. Results verified for CLIP & LLaVA across ViT-S/B/L, Sentence Transformers, & GPT-2/Qwen/Pythia backbones. **Published: EMNLP 2025 Main Conference**.
- Summer 2022 & 2023 **Research Engineer Intern at Bosch Research India**, *With Dr. Amit A. Kale & Sonam Singh*.
 - Developed an interpretable failure-discovery system** for semantic segmentation models for industry-scale autonomous driving data, **achieving 95% precision** in automatically detecting human-coherent errors via clustering. **Technical report: arXiv:2411.10845**.
 - Spearheaded the development of multi-modal image retrieval pipelines using transformers for interpretable evaluation of weather effects on camera feeds for autonomous vehicles.
- Summer 2022 & 2023 **Researcher at Trusted AI Lab IITJ**, *With Prof. Mayank Vatsa & Prof. Richa Singh*.
 - Designed a compositional negation benchmark & fine-tuned CLIP on custom objectives to achieve **SOTA accuracy on negation understanding** and $\approx 4\%$ **mean performance improvement** across diverse retrieval tasks. **Published: WACV 2025 (50 citations)**.
 - Systematically analysed StyleGAN2 latent spaces to profile identity leakage via dual-space identity invariance to inform privacy-preserving generative AI guidelines. **Published: WACV 2024**.

Technical Skills

- Machine learning**: PyTorch, TorchTitan, JAX, FLAX, Haiku, Optax, Huggingface
- Distributed training & HPC**: Slurm, Megatron-LM, DeepSpeed, Data/Tensor Parallelism
- Agentic tools**: dsPy, n8n, LangGraph, LangChain, Claude Code, OpenCode
- SWE tools**: Git, GitHub, CI/CD, testing-driven development, Selenium, Prometheus
- Programming languages**: Python, C++, Bash, JavaScript, Dart

Fellowships, Achievements, & Academic Service

- Konrad Zuse School (ELIZA) Fellowship** for Master's students 2025-2026
- Konrad Zuse School (ELIZA) Fellowship** for Master's students 2024-2025
- All-India-Rank 6428 in JEE Mains 2020, All-India-Rank 3214 in JEE Advanced 2020**
- Reviewer**: NeurIPs'26, CVPR'26, ECCV'26.